

Notes: Kelly, Pastor, and Veronesi (JF, 2016)

The Price of Political Uncertainty: Theory and Evidence from the Option Market

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
2.1	Contribution relative to policy uncertainty literature	3
2.2	Contribution relative to option-market research	3
2.3	Contribution relative to political economy asset pricing	4
2.4	Contribution relative to international finance	4
3	Data and measurement	4
3.1	Option data and country sample	4
3.2	Political events	5
3.3	Treatment and control windows	5
3.4	Variance risk premium and skew measures	5
3.5	Political uncertainty and economic conditions	6
4	Theory and hypotheses	6
4.1	Pastor-Veronesi policy-choice model	6
4.2	Stock price jumps and volatility	6
4.3	Put option pricing implication	6
4.4	Hypotheses	7
5	Empirical specifications	7
5.1	Mean-difference tests	7
5.2	Economic conditions regressions	7
5.3	Election uncertainty regressions	7
5.4	Placebo-event tests	7
5.5	Spillover and FX regressions	8
6	Identification and empirical design	8
6.1	Why event windows help	8
6.2	Main threats to identification	8
6.3	Why options are especially informative	8
7	Tables and figures	9
7.1	Table I and Figure 1: option sample and event-window construction	9
7.2	Table II and Table III: political-event sample and mean differences	9
7.3	Tables IV–VI: option premia and economic conditions	10
7.4	Figures 3–4: scatter evidence for state dependence	11
7.5	Table VII and Figure 5: election uncertainty	12
7.6	Table VIII and Table XI: placebo and spillover evidence	13
7.7	Figures 6–7: model-implied probabilities, jumps, and volatility	14

7.8	Figures 8–9: option-pricing model and international spillovers	15
8	Short summary	16
9	Conclusion	16
9.1	Core conclusion	16
9.2	Economic intuition	17
9.3	Incremental lesson	17
9.4	Takeaway	17

1 Big picture

- **Question.** How much are investors willing to pay for protection against political uncertainty, and what types of political-event risks are priced in equity index options?
- **Core idea.** Options whose lives span a political event insure against price risk, variance risk, and tail risk generated by that event. Their prices therefore reveal the market value of political uncertainty.
- **Main events.** The paper studies national elections and global summits. These events have known dates, are plausibly exogenous to option-pricing windows, and can trigger major policy or political-regime changes.
- **Identification design.** For each event, the paper compares an option whose expiration window spans the event with neighboring options on the same country index whose lives do not span the event.
- **Key treatment-control logic.** If a political event creates priced risk, the option spanning the event should be more expensive than adjacent options with similar maturity and the same underlying index.
- **Three option-market variables.** The paper studies implied volatility difference IVD, variance risk premium difference VRPD, and implied volatility slope difference SlopeD.
- **Main finding.** All three variables are higher for options spanning political events. ATM options spanning political events are about 5.1% more expensive than comparable control options.
- **Tail-risk finding.** The price premium is larger for out-of-the-money put options: about 9.4% for 5% OTM puts and 15.8% for 10% OTM puts.
- **State dependence.** Political-event option premia are higher in weaker economic conditions. Protection against political risk is more valuable when the economy is weak.
- **Uncertainty dependence.** For elections, closer races have larger option premia. Higher election uncertainty raises IVD and VRPD.
- **Spillover result.** Political uncertainty spills across borders. Foreign equity options and FX options reflect uncertainty about major political events elsewhere.
- **Takeaway.** Political uncertainty is priced in option markets in ways predicted by a model of government policy choice and learning.

2 Incremental contribution of the paper

2.1 Contribution relative to policy uncertainty literature

- Earlier work measures policy or political uncertainty using news, forecasts, or macro volatility proxies.
- Those proxies can be contaminated by general macroeconomic uncertainty.
- This paper isolates political uncertainty using event-date variation around elections and summits.
- The empirical design compares options on the same country index across nearby maturities, making the test closer to a narrow event study.

Interpretation: The paper’s key incremental contribution is to price political uncertainty directly from market prices of insurance, rather than inferring its importance from real outcomes, macro aggregates, or textual uncertainty measures.

2.2 Contribution relative to option-market research

- Option research often studies earnings announcements, macro announcements, disaster risk, or volatility risk premia.

- This paper shows that political events create economically large premia in implied volatility, variance risk premia, and volatility skews.
- It uses multiple strike prices to separate price risk, variance risk, and tail risk.
- It shows that the option market prices not only the mean effect of politics on stock prices but also uncertainty about volatility and the left tail.

Interpretation: The paper expands the set of priced option-market event risks to include political events and provides a template for studying event-specific uncertainty using option maturities.

2.3 Contribution relative to political economy asset pricing

- The paper links empirical option prices to the Pastor-Veronesi model of government policy choice.
- The model predicts that political risk premia are higher when the economy is weak and when policy outcomes are uncertain.
- The empirical results confirm these comparative statics for elections and summits.
- The paper also extends the model to option pricing and derives closed-form put option prices and jump risk premia.

Interpretation: The paper is not just an event study. It gives a structural asset-pricing interpretation of why political uncertainty should be priced more in weak economic states.

2.4 Contribution relative to international finance

- The sample spans 20 countries and covers both domestic political events and globally relevant events.
- The paper documents cross-country spillovers from Greek elections, European summits, G8/G20 meetings, and U.S. elections.
- It connects equity option premia and FX option premia to the same political uncertainty channel.

Interpretation: Political risk is not purely local. The paper documents that politically important events can affect option prices across countries and across asset classes.

3 Data and measurement

3.1 Option data and country sample

- The option data cover 20 countries.
- The underlying assets are major stock indexes, such as S&P 500, FTSE 100, DAX, CAC 40, Nikkei 225, and other leading country indexes or MSCI country ETF options.
- Most data come from OptionMetrics, while pre-1996 U.S. data come from Market Data Express.
- The sample period differs by country depending on option availability, with the U.S. series starting in 1990 and most international samples starting in the 2000s.

Interpretation: Country-level option data let the authors examine event-specific insurance prices in the equity market of the country most directly exposed to the event.

3.2 Political events

- The sample contains 271 political-event observations.
- Elections include parliamentary and presidential elections.
- Summits include Euro summits and G8/G20 summits.
- Events close together in calendar time are combined to avoid double-counting when their option windows share the same post-event expiration date.
- The paper treats the 2012 Greek elections as relevant for all European countries because of their importance for eurozone survival.

Interpretation: Combining nearby events ensures that each option-window observation corresponds to the set of political risks priced in that window.

3.3 Treatment and control windows

Let τ be the political-event date. Let a , b , and c be option expiration dates such that

$$a < \tau < b < c. \quad (1)$$

The treatment option expires at b , so its life spans the political event. Control options expire at a and c , so their lives do not span the same event.

For event τ , the implied volatility difference is

$$\text{IVD}_\tau = \overline{IV}_b - \frac{1}{2} (\overline{IV}_a + \overline{IV}_c). \quad (2)$$

Interpretation: This differencing removes country-level volatility differences and slow-moving local volatility trends, leaving the incremental option price associated with the political event.

3.4 Variance risk premium and skew measures

The variance risk premium difference is

$$\text{VRPD}_\tau = \overline{VRP}_b - \frac{1}{2} (\overline{VRP}_a + \overline{VRP}_c). \quad (3)$$

The implied volatility slope difference is

$$\text{SlopeD}_\tau = \overline{Slope}_b - \frac{1}{2} (\overline{Slope}_a + \overline{Slope}_c). \quad (4)$$

- IVD captures the extra price of ATM option protection against price risk.
- VRPD captures the extra variance risk premium embedded in event-spanning options.
- SlopeD captures the relative price of OTM put protection, hence tail-risk insurance.

Interpretation: The paper does not rely on one option statistic. It studies three related but distinct dimensions of political-event risk.

3.5 Political uncertainty and economic conditions

For elections, political uncertainty is measured as the negative of the poll spread:

$$\text{UNC} = -|\text{vote share of leading candidate or party} - \text{vote share of runner-up}|. \quad (5)$$

A closer election has a smaller margin and therefore higher UNC.

Economic conditions are measured by:

- country stock market return MKT;
- real GDP growth GDP;
- IMF GDP growth forecast FST;
- OECD Composite Leading Indicator CLI.

Interpretation: The model predicts stronger option premia in weaker economic conditions and when political outcomes are more uncertain.

4 Theory and hypotheses

4.1 Pastor-Veronesi policy-choice model

The paper uses the Pastor-Veronesi framework in which the government chooses between keeping the current policy and adopting one of several new policies. Investors are uncertain about policy choices and learn about policy costs and economic conditions.

In the election interpretation, voters choose a government or policy direction. Investors are uncertain about the election outcome and the effect of the new political regime on future cash flows and discount rates.

Interpretation: Political uncertainty is uncertainty about future policy. Elections and summits are specific dates on which that uncertainty is partly resolved.

4.2 Stock price jumps and volatility

When a policy decision is announced at time τ , stock prices can jump:

$$\frac{dM_t}{M_t} = \mu_M(S_t)dt + \sigma_M(S_t)dW_{M,t} + J_{M,t}\mathbf{1}_{t=\tau}. \quad (6)$$

The jump $J_{M,t}$ reflects surprise about the policy outcome. It depends on the state of the economy and political uncertainty.

Interpretation: Option prices are useful because they insure exactly against these jumps and volatility changes around the political event.

4.3 Put option pricing implication

For a European put option expiring after the political event, the model implies that the option price is a probability-weighted average of option values conditional on possible policy outcomes:

$$\text{Put}(S_t, m, K) = \sum_j p_{j,t} BS(S_t(1 + J_j), m, K, \sigma_j), \quad (7)$$

where $p_{j,t}$ is the probability of policy outcome j and $BS(\cdot)$ is the Black-Scholes-style conditional option value.

Interpretation: Political-event options are expensive because they insure against multiple possible policy outcomes, including downside jumps and volatility changes.

4.4 Hypotheses

1. **Political events raise option premia.** Event-spanning options should have higher IVD, VRPD, and SlopeD.
2. **Economic conditions matter.** The premia should be higher when economic conditions are weak.
3. **Outcome uncertainty matters.** For elections, closer races should have higher IVD, VRPD, and possibly SlopeD.
4. **Spillovers exist.** Major political events should affect option prices in related countries and FX markets.

5 Empirical specifications

5.1 Mean-difference tests

The first tests compare the average values of the event-window measures:

$$E[IVD_\tau], \quad E[VRPD_\tau], \quad E[SlopeD_\tau]. \quad (8)$$

Positive means imply that options expiring after political events are more expensive than neighboring control options.

Interpretation: This is the most direct test of whether the event itself is priced in the option market.

5.2 Economic conditions regressions

The paper estimates regressions of the form

$$Y_\tau = a + b \text{ECON}_\tau + c \text{ECON}_\tau \mathbf{1}\{\text{ECON}_\tau > 0\} + \varepsilon_\tau, \quad (9)$$

where Y is IVD, VRPD, or SlopeD.

Interpretation: The coefficient b should be negative: weaker economic conditions should be associated with larger event-spanning option premia.

5.3 Election uncertainty regressions

For election events, the paper estimates

$$Y_\tau = a + b \text{UNC}_\tau + c' \text{ECON}_\tau + \varepsilon_\tau, \quad (10)$$

where Y is one of the three option-market variables.

Interpretation: A positive b means that closer elections, which have more uncertain outcomes, command higher option premia.

5.4 Placebo-event tests

The paper generates placebo political events at random dates with the same frequency as true events. It recomputes IVD, VRPD, and SlopeD for these pseudo-events.

Interpretation: Placebo tests ask whether the results are a general property of random dates in option data. The paper finds the true-event effects are much stronger than placebo effects.

5.5 Spillover and FX regressions

The spillover regressions relate option premia in one country to political events in another country. The FX regressions test whether currency options become more expensive around political events.

Interpretation: These tests move beyond local political risk and show that political uncertainty can be priced internationally.

6 Identification and empirical design

6.1 Why event windows help

- Political-event dates are known in advance.
- Options around the event have short maturities, so the event is a major risk during the option life.
- The control options are from nearby expiration dates on the same underlying index.
- Differencing removes persistent cross-country and time-series volatility differences.

Interpretation: The design uses within-country, local-in-time comparisons to isolate the price of political-event risk.

6.2 Main threats to identification

- Other macroeconomic news could occur during the same option window.
- Political events might be correlated with economic stress.
- Some political risks may be resolved before or after the event date.
- Poll spreads may measure election uncertainty with error.
- Event combinations may include multiple risks in one option window.

Interpretation: The paper addresses these concerns with close control windows, exclusion of ultra-short maturities, placebo tests, robustness checks, and separate analyses of elections and summits.

6.3 Why options are especially informative

- Stock returns reveal realized price changes, not the ex ante price of insurance.
- Options reveal the ex ante cost of protection against price, volatility, and tail outcomes.
- Different strike prices identify the value of tail-risk protection.
- Variance risk premia identify compensation for volatility risk, not merely expected volatility.

Interpretation: Option prices are the natural asset prices for measuring the value of political uncertainty.

7 Tables and figures

7.1 Table I and Figure 1: option sample and event-window construction

Read:

- Table I lists the 20 countries, underlying option indexes, and sample periods.
- Figure 1 shows the treatment option expiring after the political event and the two control options expiring before and after the treatment expiration.
- The treatment option life spans the political event; control option lives do not.

Economic message: The design compares event-spanning insurance to neighboring non-event insurance on the same underlying market.

Table I
Option Sample

The table gives an overview of our option sample. For each country listed, column 1 reports the index underlying the country's options used in our analysis. Columns 2 and 3 report the beginning and ending dates for the available option data. All data come from OptionMetrics, except for pre-1996 U.S. data, which come from Market Data Express.

	Index	Start Date	End Date
	(1)	(2)	(3)
Australia	ASX 200	20040102	20120604
Belgium	BEL20	20020102	20120831
Brazil	MSCI Brazil	20060525	20120131
Canada	MSCI Canada	20060302	20120131
Finland	OMXH25	20020102	20120831
France	CAC 40	20030414	20120831
Germany	DAX	20020102	20120831
Italy	FTSE MIB	20061011	20120831
Japan	NIKKEI 225	20040506	20120604
Korea	Kospi	20040503	20120131
Mexico	MSCI Mexico	20071129	20120131
Netherlands	AEX	20020102	20120831
Singapore	MSCI Singapore	20091118	20120131
South Africa	MSCI South Africa	20070524	20120131
Spain	IBEX 35	20070514	20120831
Sweden	OMXS30	20070126	20120831
Switzerland	SMI	20020102	20120831
Taiwan	TAIEX	20040102	20120131
United Kingdom	FTSE 100	20020102	20120831
United States	S&P 500	19900101	20120131

Table I: option sample

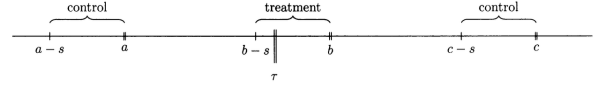


Figure 1. Timeline. Time τ is the date of the political event. Times a , b , and c are option expiration dates. For both treatment and control options, we use prices from multiple dates that are s days prior to expiration.

Figure 1: treatment and control option windows

7.2 Table II and Table III: political-event sample and mean differences

Read:

- Table II reports 271 political events: 64 elections and 216 summits, with some combined event observations.
- Table III shows positive mean differences for all three option-market variables.
- IVD averages 1.43 percentage points across all events.
- VRPD averages 1.07 after multiplying by 100.
- SlopeD averages 1.73 after multiplying by 100.

Economic message: Options spanning political events are more expensive because they insure against political-event price, variance, and tail risks.

Table II
Number of Political Events

The table reports the number of political events in our sample by event type and country. These are events for which option data are available so that it is possible to calculate *IVD* in equation (1). Totals represent the size of a set union, so they may be smaller than the sum of the subset sizes. For example, parliamentary and presidential elections may take place on the same date, in which case they are counted only once in the total. In addition, we combine into a single observation all country events that are so close together in calendar time that they are followed by the same first option expiration date (i.e., the same date *b* in Figure 1). Euro summits are included as observations for all European countries; G8 summits for all G8 countries, and G20 summits for all G20 countries. The 2012 Greek elections are included as observations for all European countries.

	Elections				Summits		
	Total	Total	Parl.	Pres.	Total	Euro	G8/G20
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
All	271	64	57	14	216	170	74
Australia	6	1	1	0	5	0	5
Belgium	13	2	2	0	11	11	0
Brazil	9	4	2	4	6	0	6
Canada	7	2	2	0	6	0	6
Finland	1	0	0	0	1	1	0
France	27	6	4	2	21	21	7
Germany	25	5	5	0	21	21	7
Italy	24	3	3	0	21	21	7
Japan	10	4	4	0	6	0	6
Korea	8	2	1	1	6	0	6
Mexico	7	1	1	0	6	0	6
Netherlands	22	3	3	0	19	19	0
Singapore	2	2	1	1	0	0	0
South Africa	6	1	1	0	5	0	5
Spain	20	4	4	0	17	17	0
Sweden	19	2	2	0	18	18	0
Switzerland	24	5	5	0	20	20	0
Taiwan	2	2	1	1	0	0	0
United Kingdom	24	4	4	0	21	21	7
United States	15	11	11	5	6	0	6

Table II: political events

Table III
Implied Volatility, Variance Risk Premium, and Slope: Mean Differences

Column 1 reports the average value of *IVD*, the implied volatility difference from equation (1), across all political events. Columns 2 through 5 report difference between the average values of *IVD* in weak economic conditions (*ECOV* < 0) and strong economic conditions (*ECOV* ≥ 0). The remaining columns report analogous results for *VRPD*, the variance risk premium difference from equation (4), and *SlopeD*, the implied volatility slope difference from equation (6). We use four measures of economic conditions: the country's stock market index return (*MKT*), real GDP growth (*GDP*), the IMF's GDP growth forecast (*FST*), and the OECD's Composite Leading Indicator (*CLI*). We standardize *GDP*, *FST*, and *CLI* to zero means within each country using data from 1990 through 2012. *IVD* is in percent per year. *VRPD* is in decimals per year. Both *VRPD* and *SlopeD* are multiplied by 100. The *t*-statistics, reported in parentheses, use standard errors with two-way clustering at the month and country levels. Panel A uses all observations, while Panels B and C report results for election and summit events separately.

	Implied Volatility (<i>IVD</i>)					Variance Risk Premium (<i>VRPD</i>)					Implied Volatility Slope (<i>SlopeD</i>)				
	Weak Minus Strong Economy					Weak Minus Strong Economy					Weak Minus Strong Economy				
	All	<i>MKT</i>	<i>GDP</i>	<i>FST</i>	<i>CLI</i>	All	<i>MKT</i>	<i>GDP</i>	<i>FST</i>	<i>CLI</i>	All	<i>MKT</i>	<i>GDP</i>	<i>FST</i>	<i>CLI</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Panel A: All Political Events															
Mean	1.43	2.57	1.94	2.22	3.00	1.07	3.02	2.07	2.55	3.05	1.73	3.36	2.11	2.66	3.02
Obs.	(4.43)	(3.79)	(3.34)	(3.78)	(4.61)	(2.61)	(3.51)	(2.80)	(3.54)	(3.63)	(3.59)	(3.11)	(2.52)	(3.08)	(2.97)
	271	271	271	266	267	271	271	271	266	267	238	238	238	233	236
Panel B: Elections Only															
Mean	1.63	2.63	1.73	2.51	2.36	1.30	2.46	1.07	2.45	1.26	1.14	3.56	1.14	1.96	1.38
Obs.	(3.13)	(2.73)	(1.78)	(2.34)	(2.39)	(2.59)	(2.62)	(1.11)	(2.30)	(1.25)	(2.08)	(3.69)	(1.08)	(1.71)	(1.28)
	64	64	64	59	60	64	64	64	59	60	55	55	55	50	53
Panel C: Summits Only															
Mean	1.42	2.68	2.13	2.40	3.25	1.07	3.39	2.58	2.97	3.75	1.84	3.43	2.53	2.87	3.46
Obs.	(3.76)	(2.27)	(3.17)	(3.56)	(4.30)	(2.15)	(3.15)	(2.92)	(3.49)	(3.74)	(3.16)	(2.88)	(2.54)	(2.83)	(2.80)
	216	216	216	216	216	216	216	216	216	216	191	191	191	191	191

Table III: mean differences

7.3 Tables IV–VI: option premia and economic conditions

Read:

- Table IV shows that *IVD* is negatively related to economic conditions.
- Table V shows the same negative relation for *VRPD*.
- Table VI shows that *SlopeD* is also generally higher in weak economic conditions.
- The state dependence appears across stock market returns, GDP growth, IMF forecasts, and OECD leading indicators.

Economic message: Political uncertainty is more valuable when the economy is weak. Investors pay more for protection against political risk when policy mistakes or adverse outcomes are more damaging.

Table IV
Implied Volatility Difference and Economic Conditions

The table reports the slope coefficients from the regressions of IVD , the implied volatility difference from equation (1), on four different measures of economic conditions ($ECON$). For each of the four measures, we estimate a simple regression of IVD on $ECON$ as well as a piecewise linear regression of IVD on both $ECON$ and $ECON$ times an indicator for $ECON$ being positive. The four measures of economic conditions are the country's stock market index return (MKT), real GDP growth (GDP), the IMF's GDP growth forecast (FST), and the OECD's Composite Leading Indicator (CLI). We standardize GDP , FST , and CLI to zero mean within each country using data from 1990 through 2012. We then scale all four $ECON$ variables to unit variance within each regression. IVD is in percent per year. The t -statistics, reported in parentheses, use standard errors with two-way clustering at the month and country levels. Panel A uses all observations, while Panels B and C report results for election and summit events separately.

	Measure of Economic Conditions							
	MKT		GDP		FST		CLI	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: All Political Events								
$ECON$	-2.42 (-5.34)	-3.58 (-5.10)	-2.01 (-4.82)	-2.71 (-5.01)	-0.83 (-2.55)	-0.63 (-1.37)	-1.39 (-3.85)	-1.47 (-2.74)
$ECON \times 1_{(ECON > 0)}$		3.77 (3.33)		4.02 (3.85)		-0.96 (-0.87)		0.26 (0.28)
R^2	0.21	0.25	0.14	0.18	0.02	0.03	0.07	0.07
Obs.	271	271	271	271	266	266	267	267
Panel B: Elections Only								
$ECON$	-1.72 (-3.49)	-2.60 (-3.51)	-1.56 (-3.03)	-2.39 (-4.63)	-1.69 (-3.00)	-2.01 (-2.09)	-1.34 (-2.15)	-1.10 (-0.86)
$ECON \times 1_{(ECON > 0)}$		2.69 (1.62)		3.26 (2.75)		1.18 (0.61)		-0.57 (-0.34)
R^2	0.17	0.20	0.14	0.18	0.15	0.16	0.12	0.13
Obs.	64	64	64	64	59	59	60	60
Panel C: Summits Only								
$ECON$	-2.68 (-5.04)	-3.93 (-4.86)	-2.27 (-4.84)	-3.02 (-5.08)	-0.77 (-2.10)	-0.54 (-1.10)	-1.53 (-3.91)	-1.88 (-3.18)
$ECON \times 1_{(ECON > 0)}$		4.16 (3.24)		4.61 (3.89)		-1.26 (-0.95)		1.23 (1.04)
R^2	0.23	0.28	0.17	0.20	0.02	0.02	0.08	0.08
Obs.	216	216	216	216	216	216	216	216

Table IV: implied volatility difference and conditions

Table VI
Implied Volatility Slope Difference and Economic Conditions

The table reports the slope coefficients from the regressions of $SlopeD$, the implied volatility slope difference from equation (6), on four different measures of economic conditions ($ECON$). For each of the four measures, we estimate a simple regression of $SlopeD$ on $ECON$ as well as a piecewise linear regression of $SlopeD$ on both $ECON$ and $ECON$ times an indicator for $ECON$ being positive. The four measures of economic conditions are the country's stock market index return (MKT), real GDP growth (GDP), the IMF's GDP growth forecast (FST), and the OECD's Composite Leading Indicator (CLI). We standardize GDP , FST , and CLI to zero mean within each country using data from 1990 through 2012. We then scale all four $ECON$ variables to unit variance within each regression. $SlopeD$ is multiplied by 100. The t -statistics, reported in parentheses, use standard errors with two-way clustering at the month and country levels. Panel A uses all observations, while Panels B and C report results for election and summit events separately.

	Measure of Economic Conditions							
	MKT		GDP		FST		CLI	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: All Political Events								
$ECON$	-3.19 (-3.82)	-5.37 (-4.21)	-2.20 (-3.02)	-2.77 (-2.87)	-0.78 (-1.66)	-0.32 (-0.48)	-1.62 (-2.47)	-1.96 (-1.89)
$ECON \times 1_{(ECON > 0)}$		6.86 (3.62)		3.11 (1.83)		-2.15 (-1.34)		1.08 (0.66)
R^2	0.19	0.26	0.09	0.10	0.01	0.02	0.05	0.05
Obs.	238	238	238	238	233	233	236	236
Panel B: Elections Only								
$ECON$	-2.11 (-5.26)	-1.70 (-2.13)	-1.21 (-2.22)	-2.34 (-3.50)	-0.31 (-0.42)	-0.18 (-0.16)	-0.98 (-1.70)	-1.05 (-0.81)
$ECON \times 1_{(ECON > 0)}$		-1.17 (-0.77)		3.38 (2.13)		-0.48 (-0.19)		0.12 (0.07)
R^2	0.27	0.28	0.09	0.15	0.01	0.01	0.06	0.06
Obs.	55	55	55	55	50	50	53	53
Panel C: Summits Only								
$ECON$	-3.54 (-3.58)	-6.15 (-4.28)	-2.49 (-3.01)	-3.01 (-2.86)	-0.88 (-1.66)	-0.28 (-0.38)	-1.78 (-2.41)	-2.20 (-1.88)
$ECON \times 1_{(ECON > 0)}$		8.42 (3.95)		3.13 (1.63)		-3.10 (-1.68)		1.41 (0.69)
R^2	0.19	0.28	0.10	0.10	0.01	0.02	0.05	0.05
Obs.	191	191	191	191	191	191	191	191

Table VI: slope and conditions

Table V
Variance Risk Premium Difference and Economic Conditions

The table reports the slope coefficients from the regressions of $VRPD$, the variance risk premium difference from equation (4), on four different measures of economic conditions ($ECON$). For each of the four measures, we estimate a simple regression of $VRPD$ on $ECON$ as well as a piecewise linear regression of $VRPD$ on both $ECON$ and $ECON$ times an indicator for $ECON$ being positive. The four measures of economic conditions are the country's stock market index return (MKT), real GDP growth (GDP), the IMF's GDP growth forecast (FST), and the OECD's Composite Leading Indicator (CLI). We standardize GDP , FST , and CLI to zero mean within each country using data from 1990 through 2012. We then scale all four $ECON$ variables to unit variance within each regression. $VRPD$ is annualized and multiplied by 100. The t -statistics, reported in parentheses, use standard errors with two-way clustering at the month and country levels. Panel A uses all observations, while Panels B and C report results for election and summit events separately.

	Measure of Economic Conditions							
	MKT		GDP		FST		CLI	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: All Political Events								
$ECON$	-2.98 (-4.56)	-4.55 (-4.45)	-1.97 (-3.52)	-2.43 (-3.40)	-1.32 (-3.42)	-1.40 (-2.50)	-1.72 (-3.38)	-2.23 (-2.88)
$ECON \times 1_{(ECON > 0)}$		5.10 (3.41)		2.62 (1.67)		0.35 (0.27)		1.75 (1.44)
R^2	0.19	0.24	0.08	0.09	0.04	0.04	0.06	0.07
Obs.	271	271	271	271	266	266	267	267
Panel B: Elections Only								
$ECON$	-1.62 (-2.34)	-2.22 (-1.90)	-1.32 (-1.78)	-2.79 (-3.80)	-1.78 (-2.69)	-1.76 (-1.70)	-1.18 (-1.51)	-1.79 (-1.18)
$ECON \times 1_{(ECON > 0)}$		1.86 (0.92)		5.79 (3.73)		-0.09 (-0.05)		1.40 (0.70)
R^2	0.16	0.18	0.11	0.25	0.18	0.18	0.09	0.10
Obs.	64	64	64	64	59	59	60	60
Panel C: Summits Only								
$ECON$	-3.45 (-4.52)	-5.27 (-4.53)	-2.32 (-3.67)	-2.67 (-3.39)	-1.40 (-3.13)	-1.53 (-2.48)	-2.01 (-3.56)	-2.69 (-3.08)
$ECON \times 1_{(ECON > 0)}$		6.07 (3.59)		2.17 (1.18)		0.71 (0.46)		2.37 (1.56)
R^2	0.22	0.28	0.10	0.10	0.04	0.04	0.07	0.08
Obs.	216	216	216	216	216	216	216	216

Table V: variance risk premium and conditions

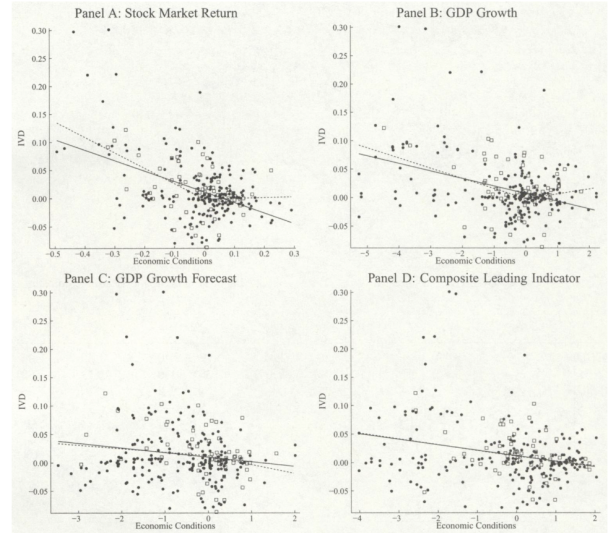


Figure 2. Implied volatility difference versus economic conditions: data. The figure plots IVD , the annualized implied volatility difference from equation (1), against four measures of economic conditions: the country's stock market index return (MKT ; Panel A), real GDP growth (GDP ; Panel B), the IMF's GDP growth forecast (FST ; Panel C), and the OECD's Composite Leading Indicator (CLI ; Panel D). We standardize GDP , FST , and CLI to zero mean and unit variance within each country using data from 1990 through 2012. Election observations are marked as squares; summit observations as circles. The solid line is the line of best fit from a simple linear regression. The dashed line is from the piecewise linear regression with a kink at zero.

Figure 2: IVD and economic conditions

7.4 Figures 3–4: scatter evidence for state dependence

Read:

- Figure 3 plots $VRPD$ against economic conditions.

- Figure 4 plots SlopeD against economic conditions.
- Both figures show more dispersion and higher premia in weak conditions.
- The patterns are not driven by a few visible outliers.

Economic message: The scatter plots reinforce the regression evidence: political-event protection is more valuable when macroeconomic conditions are weak.

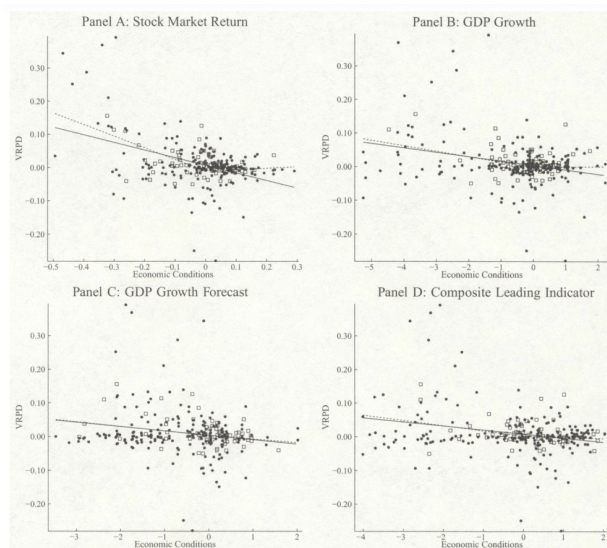


Figure 3. Variance risk premium difference versus economic conditions: data. The figure plots $VRPD$, the annualized variance risk premium difference from equation (4), against four measures of economic conditions: the country's stock market index return (MKT ; Panel A), real GDP growth (GDP ; Panel B), the IMF's GDP growth forecast (FST ; Panel C), and the OECD's Composite Leading Indicator (CLI ; Panel D). We standardize GDP , FST , and CLI to zero mean and unit variance within each country using data from 1990 through 2012. Election observations are marked as squares; summit observations as circles. The solid line is the line of best fit from a simple linear regression. The dashed line is from the piecewise linear regression with a kink at zero.

Figure 3: VRPD and economic conditions

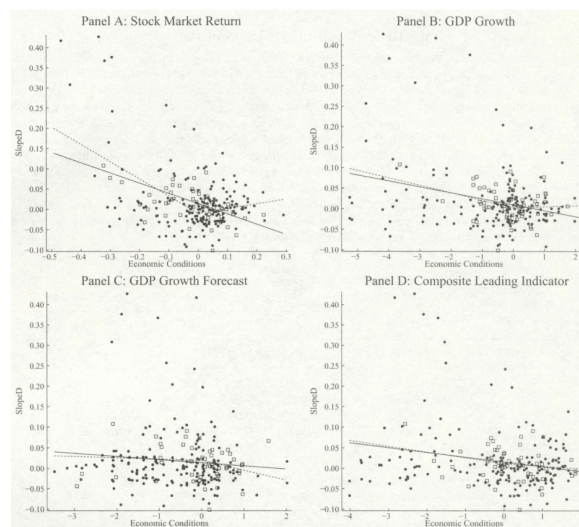


Figure 4. Implied volatility slope difference versus economic conditions: data. The figure plots $SlopeD$, the implied volatility slope difference from equation (6), against four measures of economic conditions: the country's stock market index return (MKT ; Panel A), real GDP growth (GDP ; Panel B), the IMF's GDP growth forecast (FST ; Panel C), and the OECD's Composite Leading Indicator (CLI ; Panel D). We standardize GDP , FST , and CLI to zero mean and unit variance within each country using data from 1990 through 2012. Election observations are marked as squares; summit observations as circles. The solid line is the line of best fit from a simple linear regression. The dashed line is from the piecewise linear regression with a kink at zero.

Figure 4: SlopeD and economic conditions

7.5 Table VII and Figure 5: election uncertainty

Read:

- Table VII relates option-market variables to election uncertainty UNC .
- A one-standard-deviation increase in UNC increases IVD by 1.87 percentage points in the simple regression.
- $VRPD$ also rises significantly with UNC .
- The evidence for $SlopeD$ is positive but weaker.
- Figure 5 plots the data underlying the simple uncertainty regressions.

Economic message: Closer elections command larger option premia. Markets price not only the existence of elections but also the uncertainty about their outcomes.

Table VII
The Role of Election Uncertainty

Panel A reports the slope coefficients from the regressions of IVD , the implied volatility difference from equation (1), on our measure of political uncertainty (UNC). Panel B replaces IVD with $VRPD$, the variance risk premium difference from equation (4), while Panel C replaces it with $SlopeD$, the slope difference from equation (6). The uncertainty measure UNC , which is defined as the negative of the election poll spread, proxies for uncertainty about the election outcome. We estimate a simple regression on UNC as well as multiple regressions that include controls for four different measures of economic conditions ($ECON$), in both linear and piecewise linear specifications. The four measures are the country's stock market index return (MKT), real GDP growth (GDP), the IMF's GDP growth forecast (FST), and the OECD's Composite Leading Indicator (CLI). Both IVD and $VRPD$ are annualized and multiplied by 100. $SlopeD$ is also multiplied by 100. We scale UNC to unit variance within each regression. The t -statistics, reported in parentheses, use standard errors with two-way clustering at the month and country levels.

	Measure of Economic Conditions								
	<i>MKT</i>		<i>GDP</i>		<i>FST</i>		<i>CLI</i>		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Implied Volatility (<i>IVD</i>)									
<i>UNC</i>	1.87 (5.39)	1.64 (4.52)	1.58 (4.43)	1.80 (4.93)	1.82 (4.49)	1.68 (4.11)	1.69 (4.11)	1.44 (3.21)	1.45 (3.14)
<i>ECON</i>		-1.46 (-2.98)	-2.17 (-2.85)	-1.47 (-3.27)	-2.34 (-5.57)	-1.36 (-2.45)	-1.72 (-1.89)	-1.52 (-3.26)	-1.61 (-1.77)
<i>ECON</i> × 1 _(ECON>0)			2.14 (1.39)		3.41 (2.65)		1.34 (0.80)		0.22 (0.17)
<i>R</i> ²	0.20	0.31	0.34	0.32	0.37	0.29	0.30	0.26	0.26
Obs.	64	64	64	64	64	59	59	60	60
Panel B: Variance Risk Premium (<i>VRPD</i>)									
<i>UNC</i>	1.06 (3.48)	0.82 (2.39)	0.78 (2.32)	0.99 (2.54)	1.03 (2.31)	0.74 (1.91)	0.74 (1.89)	0.89 (1.75)	1.00 (1.76)
<i>ECON</i>		-1.49 (-2.10)	-2.01 (-1.66)	-1.27 (-1.75)	-2.76 (-3.90)	-1.64 (-2.38)	-1.63 (-1.58)	-1.29 (-1.80)	-2.15 (-1.66)
<i>ECON</i> × 1 _(ECON>0)			1.59 (0.79)		5.87 (3.69)		-0.02 (-0.01)		1.95 (1.09)
<i>R</i> ²	0.07	0.20	0.21	0.17	0.32	0.21	0.21	0.14	0.17
Obs.	64	64	64	64	64	59	59	60	60
Panel C: Implied Volatility Slope (<i>SlopeD</i>)									
<i>UNC</i>	0.25 (0.59)	-0.11 (-0.33)	-0.14 (-0.39)	0.07 (0.17)	0.11 (0.27)	0.25 (0.50)	0.24 (0.47)	0.23 (0.56)	0.24 (0.57)
<i>ECON</i>		-2.13 (-5.08)	-1.71 (-2.12)	-1.20 (-2.16)	-2.33 (-3.39)	-0.28 (-0.36)	-0.17 (-0.15)	-0.98 (-1.67)	-1.07 (-0.82)
<i>ECON</i> × 1 _(ECON>0)			-1.20 (-0.78)		3.40 (2.13)		-0.39 (-0.16)		0.18 (0.10)
<i>R</i> ²	0.00	0.27	0.28	0.09	0.15	0.01	0.01	0.06	0.06
Obs.	55	55	55	55	55	50	50	53	53

Table VII: election uncertainty

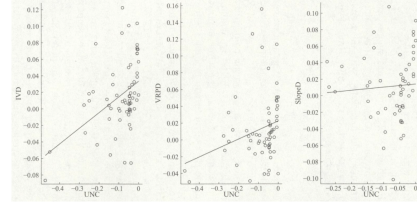


Figure 5: The role of political uncertainty: Data. The figure plots three option market quantities against uncertainty about election outcome (UNC) for all election observations in our sample. UNC is the negative of the election poll spread. The three quantities are: IVD , the annualized implied volatility difference from equation (1), $VRPD$, the annualized variance risk premium difference from equation (4), and $SlopeD$, the implied volatility slope difference from equation (6). We also show the line of best fit from a simple linear regression.

Figure 5: election uncertainty scatter plots

7.6 Table VIII and Table XI: placebo and spillover evidence

Read:

- Table VIII compares true political events to placebo pseudo-events.
- True events generate much larger IVD , $VRPD$, and $SlopeD$ than placebo dates.
- Table XI shows that spillover IVD and $VRPD$ are related to FX option premia.
- The spillover regressions support the two-country model's prediction that political risk travels across markets.

Economic message: The placebo results rule out random-date explanations. The spillover results show that political uncertainty can affect foreign equity and currency option markets.

Table VIII
Placebo Events and Mean Differences

The first column in each panel compares the average values of *IVD* (Panel A), *VRPD* (Panel B), and *SlopeD* (Panel C) from two samples: our actual data sample and a sample of placebo events. We generate 10,000 placebo samples, each including 271 placebo event dates that are randomly selected from the option samples listed in Table I. We impose that the frequency of draws from each country's sample matches the frequency of events in column 1 of Table II and that placebo events do not occur within 30 days of an actual political event in our sample. The first column reports averages of *IVD*, *VRPD*, and *SlopeD* across all placebo events, while the remaining columns report differences between the averages in weak economic conditions ($ECON < 0$) and strong economic conditions ($ECON \geq 0$). We use four measures of economic conditions: the country's stock market index return (*MKT*), real GDP growth (*GDP*), the IMF's GDP growth forecast (*FST*), and the OECD's Composite Leading Indicator (*CLI*). We standardize *GDP*, *FST*, and *CLI* to zero mean within each country using data from 1990 through 2012. *IVD* is in percent per year, *VRPD* is in decimals per year, and both *VRPD* and *SlopeD* are multiplied by 100. The reported *p*-values are the fractions of placebo estimates that exceed the actual estimate in the direction predicted by theory.

		Weak Minus Strong Economy			
	All (1)	<i>MKT</i> (2)	<i>GDP</i> (3)	<i>FST</i> (4)	<i>CLI</i> (5)
Panel A: <i>IVD</i>					
Mean (data)	1.43	2.57	1.94	2.22	3.00
Mean (pseudoevents)	-0.32	1.35	-0.63	-1.63	-1.03
Difference	1.75	1.22	2.57	3.85	4.03
<i>p</i> -Value	<0.001	0.017	<0.001	<0.001	<0.001
Obs.	271	271	271	266	267
Panel B: <i>VRPD</i>					
Mean (data)	1.07	3.02	2.07	2.55	3.05
Mean (pseudoevents)	-0.26	0.21	-0.11	-0.44	-0.61
Difference	1.33	2.81	2.18	2.99	3.66
<i>p</i> -Value	<0.001	<0.001	<0.001	<0.001	<0.001
Obs.	271	271	271	266	267
Panel C: <i>SlopeD</i>					
Mean (data)	1.73	3.26	2.11	2.66	3.02
Mean (pseudoevents)	-0.31	0.41	-0.65	0.59	-0.46
Difference	2.04	2.85	2.75	2.06	3.48
<i>p</i> -Value	<0.001	0.005	<0.001	<0.001	<0.001
Obs.	238	238	238	233	236

Table VIII: placebo events

Table XI
Relations between Spillovers and FX Options

Panel A reports slope coefficients from regressions of spillover *IVD*, denoted by *SpillIVD*, on FX *IVD*, denoted by *FXIVD*, across political events. For each political event (U.S. election or European summit), *SpillIVD* is the equity option *IVD* for the country not experiencing the event. That is, for each European summit, *SpillIVD* is the U.S. *IVD*, and for each U.S. election, *SpillIVD* is the average *IVD* across all eurozone countries. *FXIVD* for each event is *IVD* for ATM EUR/USD FX put and call options. *ECON* denotes controls for four measures of economic conditions in the country experiencing the event. The numbers in parentheses are *t*-statistics. The sample begins in January 1999 when FX option data become available. Panel B contains analogous results for *VRPD*. We do not report results for *SlopeD* because its construction requires data on FX option deltas. That data become available only in November 2010, as a result of which the regression analysis would be based on only six observations and thus unreliable.

	<i>MKT</i>		<i>GDP</i>		<i>FST</i>		<i>CLI</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: <i>IVD</i>								
<i>FXIVD</i>	1.05 (2.71)	0.97 (2.69)	0.99 (2.70)	1.09 (2.91)	1.07 (2.82)			
<i>ECON</i>		-14.97 (-2.32)	-1.11 (-2.11)	-1.69 (-1.69)	-0.89 (-1.47)			
<i>R</i> ²	0.227	0.369	0.348	0.309	0.291			
Obs.	27	27	27	27	27			
Panel B: <i>VRPD</i>								
<i>FXVRPD</i>	4.19 (2.69)	4.16 (2.74)	4.21 (2.70)	4.66 (3.14)	4.41 (2.84)			
<i>ECON</i>		-16.85 (-1.51)	-0.86 (-0.94)	-3.30 (-2.04)	-1.21 (-1.19)			
<i>R</i> ²	0.224	0.292	0.252	0.339	0.268			
Obs.	27	27	27	27	27			

Table XI: spillovers and FX options

7.7 Figures 6–7: model-implied probabilities, jumps, and volatility

Read:

- Figure 6 plots policy probabilities and stock price jumps in the Pastor-Veronesi model.
- Bad economic conditions make policy replacement more likely and make policy-induced stock price jumps larger.
- Figure 7 plots diffusive and expected stock return volatility.
- Expected volatility before the political event includes both diffusive volatility and jump risk around the event.

Economic message: The model explains why political uncertainty affects both expected price jumps and volatility, especially in weak economies.

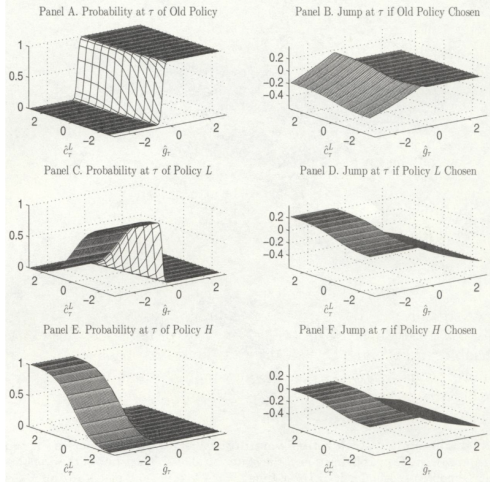


Figure 6. Model-implied policy probabilities and jumps in stock prices. The left-hand-side panels of this figure plot the adoption probabilities of the three government policies (the current or “old” policy and two potential new policies, L and H) immediately before the policy decision is made at time τ . The right-hand-side panels plot the instantaneous stock market returns conditional on the three potential policy announcements. All quantities are plotted against two state variables, $\hat{g}_t$ (the state of the economy) and $\hat{c}_t^L$ (the perceived political cost of policy L), while holding the third stochastic variable, $\hat{c}_t^H$, equal to its prior mean. The values of $\hat{g}_t$ and $\hat{c}_t^L$ are scaled by their posterior standard deviations (and thus converted into “z-scores”). All quantities are computed within the PV model for baseline parameter values.

Figure 6: policy probabilities and price jumps

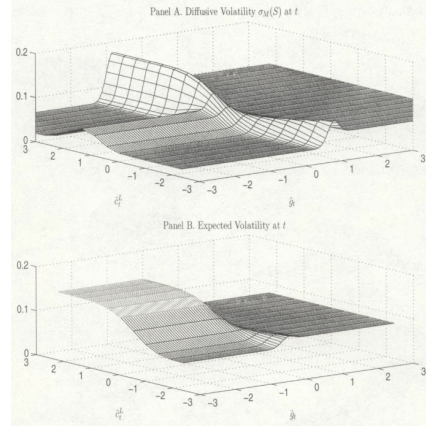


Figure 7. Model-implied stock return volatility. Panel A plots the diffusive volatility of the stock price process, $\sigma_M(S_t)$, as a function of the state variables $\hat{g}_t$ (the state of the economy) and $\hat{c}_t^L$ (the perceived political cost of policy L) as of time $t = \tau - 1/2$. Panel B plots the expected volatility of log stock returns between times $\tau - \frac{1}{2}$ and $\tau + \frac{1}{2}$, evaluated as of time $t = \tau - 1/2$. The values of $\hat{g}_t$ and $\hat{c}_t^L$ are scaled by their posterior standard deviations (and thus converted into “z-scores”). The third stochastic variable, $\hat{c}_t^H$, is set equal to its prior mean. All quantities are computed within the PV model for baseline parameter values.

Figure 7: model-implied stock return volatility

7.8 Figures 8–9: option-pricing model and international spillovers

Read:

- Figure 8 shows that model-implied implied volatility, variance risk premia, and volatility slopes are highest when economic conditions are weak and political uncertainty is high.
- Figure 9 extends the model to two countries and shows how political uncertainty in one country affects equity options in another and FX options between the countries.

Economic message: The model reproduces the paper’s main empirical patterns and shows why spillovers should exist when political risks are internationally connected.

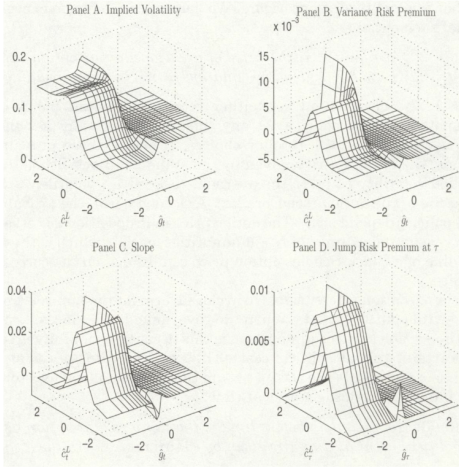


Figure 8. Option pricing implications of the PV model. This figure plots the three option market variables, implied volatility (Panel A), variance risk premium (Panel B), and slope (Panel C), as a function of the state variables $\hat{g}_t$ (the state of the economy) and $\hat{c}_t^L$ (the perceived political cost of policy L). The variables are computed as of time $t = \tau - 1/2$ for options expiring at time $\tau + 1/2$, where τ is the date of the political event. The values of $\hat{g}_t$ and $\hat{c}_t^L$ are scaled by their posterior standard deviations (and thus converted into “z-scores”). The third stochastic variable, $\hat{c}_t^H$, is set equal to its prior mean. Panel D reports the jump risk premium, or the expected instantaneous stock market return, computed immediately before time τ . All quantities are computed within the PV model for baseline parameter values.

Figure 8: model-implied option variables

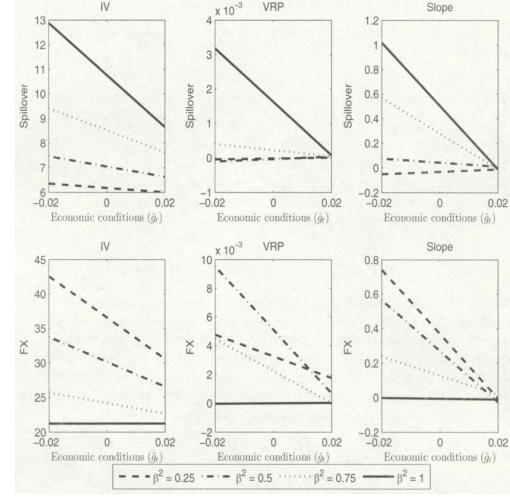


Figure 9. The effects of political uncertainty in a two-country model. This figure plots the three option market variables, implied volatility (IV), variance risk premium (VRP), and slope (Slope), against economic conditions, measured by the state variable $\hat{g}_t$. The variables are computed as of time $t = \tau - 1/2$ for options expiring at time $\tau + 1/2$, where τ is the date of the political event. Each panel plots the regression line of best fit estimated across a large number of samples simulated from the two-country extension of the PV model with $\alpha = 1$. The top three panels show spillover effects, defined as the effects of political uncertainty in the home country on equity option prices in the foreign country. The bottom three panels show the effects of political uncertainty in the home country on the prices of foreign exchange options.

Figure 9: two-country spillover model

8 Short summary

- The paper prices political uncertainty using equity index options around elections and summits.
- The empirical design compares event-spanning options with nearby control options on the same country index.
- Political events raise ATM implied volatility, variance risk premia, and implied volatility slopes.
- ATM put options spanning political events are about 5.1% more expensive.
- OTM put premiums are even larger, indicating priced tail risk.
- The political-event premium is higher in weak economic conditions.
- Closer elections, measured by poll spread, have higher option premia.
- Placebo events do not reproduce the main effects.
- Political uncertainty spills over to other countries and to FX options.
- The Pastor-Veronesi model explains these patterns through policy-choice uncertainty, learning, price jumps, and volatility risk.

9 Conclusion

9.1 Core conclusion

Kelly, Pastor, and Veronesi show that political uncertainty is priced in option markets. Options whose lives span elections and summits are more expensive because they insure investors against price risk, variance risk, and tail risk linked to political events.

9.2 Economic intuition

Political events resolve uncertainty about policy choices. In weak economic states, a bad policy outcome is more damaging and policy replacement is more likely. Investors therefore value protection more when the economy is weak and when the political outcome is uncertain. Option prices reveal this value directly.

9.3 Incremental lesson

The paper's main incremental lesson is methodological and substantive. Methodologically, short-dated options around known political-event dates isolate political risk more cleanly than broad time-series uncertainty proxies. Substantively, the paper shows that politics affects not only expected stock returns but also option-implied volatility, variance risk premia, skewness, and international spillovers.

9.4 Takeaway

Political uncertainty is an asset-pricing state variable. It affects the prices of insurance claims, is strongest in bad times, rises when elections are close, and crosses borders through equity and currency markets.